Vision Health Prediction with NHIS Data

Pritam Gupta
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
bl.en.u4cse23071@bl.students.amrita.edu

Parth Pathak
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
bl.en.u4cse23036@bl.students.amrita.edu

Pratyush Swain
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
bl.en.u4cse23043@bl.students.amrita.edu

Dr. Peeta Basa Pati
Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Bengaluru, India
bp_ peeta@blr.amrita.edu

***Abstract*—Eye health prediction is a significant problem in public healthcare, particularly for early detection and improved quality of life. In our project, we apply machine learning methods to predict categories of eye disease according to individual health and demographic information. The data is derived from the National Health Interview Survey (NHIS) – Vision and Eye Health Surveillance system. It encompasses attributes like age, gender, medical history, and self-reported vision problems. We use multiple regression and classification algorithms like Decision Tree, Random Forest, Support Vector Machines (SVM), Linear Regression, Ridge, and Lasso to test their performance. The findings reveal that machine learning can assist with vision health analysis and enhance healthcare decision-making.**

***Index Terms*—Index Terms—Machine learning, eye disease prediction, vision health, NHIS dataset, regression, classification, healthcare analytics**

I. Introduction

Vision disorders impact millions of individuals and represent a significant public health concern. Early identification of those at risk for eye disease can mitigate long-term consequences and enhance quality of life. The National Health Interview Survey (NHIS) offers accurate data gathered from a general population, including vision-related information like trouble seeing, access to eye care, and history of disease like glaucoma or cataract.

This project seeks to develop machine learning models that forecast the category of vision status or eye disease based on demographic and health inputs. We investigate models like Linear Regression, Decision Trees, Random Forest, Support Vector Regression (SVR), Ridge, and Lasso to identify which of them work best with the NHIS Vision and Eye Health Surveillance data. The forecasted categories are conditions like "difficulty seeing," "self-reported glaucoma," and "night vision problems," among others.

By assessing these models, we seek to demonstrate the ways data-driven methods may aid vision health surveillance and offer beneficial instruments for early detection and preventive measures in the healthcare system.

II. Literature Survey

[1] The research is based on the analysis of NHIS data using the sample of 2008, 2016, and 2017 to understand the relationship between social determinants of health (SDOH) and the outcomes of cataracts. The study utilizes a multivariable logistic regression analysis, which reveals that some of the main factors that predispose cataract diagnosis, vision impairment, and cataract surgery include age, unemployment, inability to cover the medical bills, lack of insurance cover, and low income. These results highlight the necessity to use the screening based on the social risks in their ophthalmological practice on a regular basis.

[2] It treats a subject addressing a global outlook, whereby data utilised by this study is based on the China Health and Retirement Longitudinal Study, where insertion of multiple machine learning models such as gradient boosting and ensemble models were used to forecast VI. Determinants like hearing impairment, self-perceived health status, pain, age, hand grip strength and depression, are successfully identified and predict the importance of advanced prediction models that identify and intervene the huge population at an early stage.

[3] The authors discuss relationships between social determinants and self-rated difficulty in seeing using NHIS-2021 data to inspect the associations of more than 30,000 adults. Female sex, LGBTQ identification, public insurance coverage, and lower education and low income are associated with higher vision difficulty, which highlights the ongoing importance of sociodemographic disparities invisual health.

[4] This cross-sectional study of children and adolescent individuals in the NHIS shows that, there are high stratifications between the vision difficulty and other healthcare affordability, public insurance, age and parent education. The research identifies the role of the social determinants of child

and household levels and requests the age-specific policy intervention in the health of the vision of young people.

[5] A comparative view on the self-reported versus examination-based estimates of the five largest surveys in US (NHIS, NHANES, ACS, BRFSS, NSCH) indicates a huge variety of prevalence statistics on VI and blindness, all of the datasets demonstrate dramatic age-related growing tendency. The study promotes the standardization as well as harmonization of vision-health related tools in national surveys.

[6] The sample size comprises 586 seniors, which is used to develop and validate a risk prediction model based on logistic regression, reaching high accuracy (AUC = 0.87). The major predictors are age, systolic blood pressure, physical health activity, diabetes, ocular disease history, and education. The findings justify the use of predictive analytics in preventative eye care.

[7] In this NHATS-based analysis, this demonstrates that being VI in older adults increases the risk of food insecurity more than twofold, which illustrates the synergistic risks of being at risk in both matters, and the authors recommend integrating services to enhance meet the combined health needs.

[8] This paper concentrating on a large sample of adults of low income group highlights a dose-response in association between VI and food insecurity. The results indicate that eye health condition is a significant determinant of other broader health indicators especially in socioeconomically marginal groups. Centers for Disease Control and Prevention (CDC), Vision and Eye Health Surveillance System (VEHSS) Surveillance System Reports Using NHIS Data, VEHSS Summary.

[9] With objective evaluation, the authors examine the 2021 National Health and Aging Trends Study to state that 27.8 percent of adult population aged 71+ in the US are visual impaired. The prevalence of vision loss is greatest in older, less educated, lower income, and non-White populations thus renewing the inequality problem and informing of the need to target public health interventions.

[10] This research examined how prevalent vision issues are among adults 71 and older in the U.S. Through the use of national health statistics, the scientists discovered that elderly individuals experience a lot of vision problems, which go untreated more often than not. The research points to a need for more monitoring and early intervention. It is in line with the belief that employing data and forecasting models such as our project will improve eye health and inform health policies. Their results also highlight the need for targeted intervention among older people to prevent avoidable loss of vision.

III. Methodology

A. Data Preparation

The dataset used in this study was loaded from an Excel file. To ensure the validity of the correlation analysis, only numerical features present in the dataset were selected, including `YearStart`, `YearEnd`, `Data_Value`, `Low_Confidence_Limit`, `High_Confidence_Limit`, `Sample_Size`, and `LocationID`. All missing values were checked and handled as necessary to avoid bias in the results.

B. Correlation Analysis and Visualization

The Pearson correlation coefficient was computed for all pairs of numerical attributes. This statistical measure quantifies the linear relationship between each feature pair in the dataset. To facilitate the interpretation of these relationships, a heatmap plot was generated using the seaborn library. The heatmap visually represents the strength and direction of the correlations, with warmer colors indicating strong positive associations and cooler colors corresponding to negative or weak relationships.

C. Dimensionality Reduction Using PCA

To address potential redundancy and multicollinearity among the features, Principal Component Analysis (PCA) was applied for dimensionality reduction. PCA transforms the original correlated variables into a new set of uncorrelated components, known as principal components, that capture the maximum possible variance in the data. Prior to applying PCA, all feature values were standardized using `StandardScaler` to ensure that each feature contributed equally to the analysis regardless of its original scale.

Two PCA configurations were implemented as part of this study:

- **Experiment A2:** The number of principal components was selected such that 99% of the total variance in the dataset was retained.
- **Experiment A3:** The number of principal components was selected to retain 95% of the total variance.

For each configuration, the cumulative explained variance ratio was calculated to determine the number of components required to achieve the specified variance threshold. Scree plots and cumulative variance plots were generated to visualize the contribution of each principal component and to validate the variance retention in both experiments.

D. Outlier Detection and Removal

After PCA transformation, outlier detection was carried out using the Isolation Forest algorithm. This method identifies anomalous observations based on the principle of random partitioning, where outliers require fewer splits to be isolated compared to normal samples. A contamination rate of 10% was set, meaning that approximately the top 10% most isolated points were removed as outliers. This step ensured that only representative, noise-free samples were used in subsequent classification tasks.

E. Model Training and Evaluation

Following data cleaning and dimensionality reduction, the dataset was divided into training (80%) and testing (20%) subsets using stratified sampling to maintain class balance. Multiple classification models were then trained on the PCA-transformed features to assess the performance impact of dimensionality reduction. The models included:

1) Logistic Regression (LR)

2) Random Forest Classifier (RF)
3) Gradient Boosting Classifier (GB)
4) Support Vector Machine (SVM)
5) Multi-Layer Perceptron (MLP)

Each model was trained on the training data and evaluated on the test set. In addition to the test accuracy, 5-fold cross-validation was conducted to estimate model stability and generalization. For each model, the mean and standard deviation of cross-validation scores were reported.

F. Performance Visualization

To facilitate interpretation and comparison of model performance, several visual analyses were conducted. Bar plots were generated to compare test accuracies and cross-validation results among all models for both variance thresholds (99% and 95%). Furthermore, the confusion matrix of the best-performing model in each experiment was visualized using a heatmap to better understand class-level prediction behavior.

G. Comparative Analysis Between 99% and 95% Variance

Finally, the two experiments (A2 and A3) were compared to analyze the trade-off between the level of dimensionality reduction and model performance. The comparison included:

- Number of principal components retained
- Percentage of dimensionality reduction
- Best-performing model in each configuration
- Differences in test accuracy and cross-validation mean scores

The results demonstrated how varying the retained variance threshold influenced both computational efficiency and predictive accuracy. Visual comparisons, including bar plots for PCA components and model accuracies, were created to summarize these findings effectively. All numerical results and figures were saved for documentation and reproducibility.

H. Feature Selection and Model Explainability Using PCA, SFS, LIME, and SHAP

The dataset was first loaded from an Excel file and preprocessed to ensure compatibility for machine learning operations. The target variable, denoted as `Response`, was label-encoded to convert categorical outputs into numerical form. Non-numeric columns were excluded, and missing values within numeric features were imputed using the mean strategy via `SimpleImputer`. This ensured that no missing data introduced bias during model training.

Two primary dimensionality and feature selection methods were explored. First, Principal Component Analysis (PCA) was applied for dimensionality reduction with retained variance levels of 99% and 95%. The feature matrix was standardized using `StandardScaler` before applying PCA to ensure equal feature contribution. A Random Forest Classifier with 100 estimators was trained and evaluated on PCA-transformed data to compute accuracy for each variance threshold.

Next, Sequential Feature Selection (SFS) was performed to identify the five most significant features. This forward selection approach iteratively selected features that maximized model performance when combined with previously chosen features. The same Random Forest Classifier was used as the base estimator to maintain consistency in comparative performance.

To enhance interpretability, post-model explanations were conducted using LIME and SHAP frameworks. LIME (Local Interpretable Model-agnostic Explanations) provided localized feature importance for individual predictions, while SHAP (SHapley Additive exPlanations) quantified global feature contributions based on cooperative game theory principles. Both explanation techniques were applied to the Random Forest model trained on the original dataset to visualize and interpret influential features affecting the model's predictions.

I. Sequential Feature Selection and Explainability Analysis

This phase focused exclusively on identifying the most influential subset of features and understanding their contribution to the model's predictive behavior. The dataset was imported from an Excel file, and preprocessing steps mirrored those of earlier experiments. The target variable `Response` was label-encoded, and all numerical attributes were retained after imputing missing values using their respective means to ensure consistency in the dataset.

A Random Forest Classifier was employed as the base model, and Sequential Feature Selection (SFS) was utilized to select the top five features that provided the highest classification accuracy. The dataset was split into 80% training and 20% testing partitions using stratified sampling to preserve class proportions. After feature selection, the classifier was retrained on the reduced feature space, and the model's accuracy on the test data was recorded.

To provide interpretable insights into model predictions, LIME and SHAP were used to generate local and global explanations, respectively. LIME explained the impact of individual features for specific instances, while SHAP provided a comprehensive view of feature importance across all samples. These visual interpretability tools aided in understanding which features most strongly influenced model decisions and validated that the selected features were meaningful within the problem domain.

IV. Results

A. Feature Correlation Findings

The resulting heatmap, shown in Fig. 1, provides an overview of the interdependencies among the selected features. Highly positive correlations were observed between `Data_Value`, `Low_Confidence_Limit`, and `High_Confidence_Limit`, indicating these variables are strongly related and may contain redundant information. The attributes `YearStart` and `YearEnd` also exhibited near-perfect positive correlation due to their direct temporal connection. In contrast, weak or negative correlations were present between certain pairs, such as `Sample_Size` and other variables, suggesting a limited linear relationship.

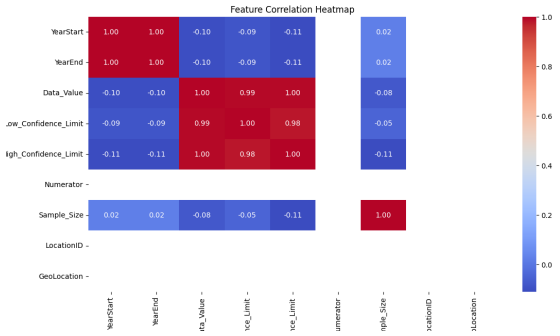


Fig. 1. Feature correlation heatmap depicting the relationships among numerical attributes.

The visualization facilitates rapid identification of highly correlated or independent features, which is valuable for further feature selection and modeling.

B. Principal Component Analysis (PCA) Results

Principal Component Analysis (PCA) was performed to identify the minimum number of components that retained most of the dataset's variance. Two configurations were evaluated — one preserving 99% of the explained variance and another preserving 95%.

For the 99% variance configuration, the dimensionality was reduced from 23 original features to 14 principal components, representing a 39.13% reduction in feature space. For the 95% configuration, only 11 components were required, achieving a 52.17% reduction while maintaining strong representational capacity.

The cumulative variance plots in Fig. 2 and Fig. 3 illustrate how the principal components capture increasing amounts of variance. The first few components account for the majority of the dataset's information, validating the high compressibility of the data without significant information loss.

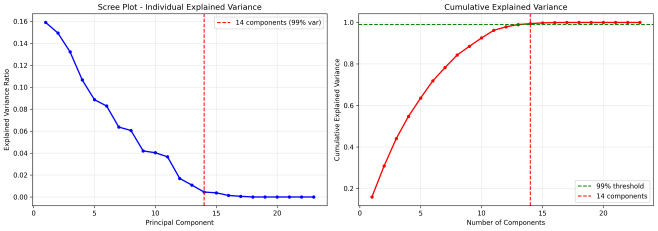


Fig. 2. Explained variance ratio across principal components (99% variance configuration).

C. Model Performance Evaluation

Five models were trained and evaluated under both PCA configurations: Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine (SVM), and a Neural Network. Each model's performance was assessed using test accuracy and five-fold cross-validation mean scores.

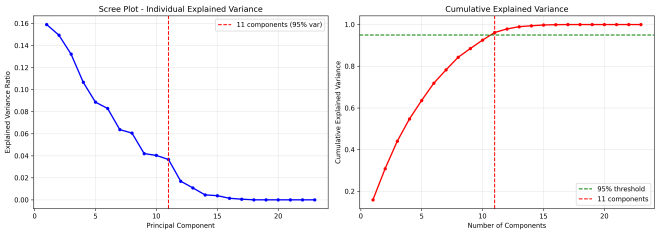


Fig. 3. Explained variance ratio across principal components (95% variance configuration).

Under the 99% variance configuration, all models achieved exceptionally high accuracy, with the Neural Network attaining a perfect score of 1.0000 on both test and cross-validation sets. The Random Forest and Logistic Regression models also demonstrated near-perfect results, indicating the effectiveness of the preprocessed and reduced feature space.

For the 95% variance configuration, the Neural Network again achieved a perfect accuracy score, confirming its robustness even with fewer principal components. However, a slight performance decline was observed for ensemble models such as Random Forest and Gradient Boosting, reflecting minor sensitivity to reduced feature dimensionality.

D. Comparative Analysis of PCA Configurations

To examine the impact of dimensionality reduction on classification performance, a comparative study was conducted between the 99% and 95% explained variance configurations. Fig. 4 summarizes the accuracy differences across all models. While the Neural Network maintained identical performance in both configurations, other models showed marginal drops in accuracy, particularly Gradient Boosting (from 0.9997 to 0.9636).

This observation indicates that while aggressive dimensionality reduction can slightly affect tree-based models, neural models remain largely invariant to moderate reductions in input dimensionality.

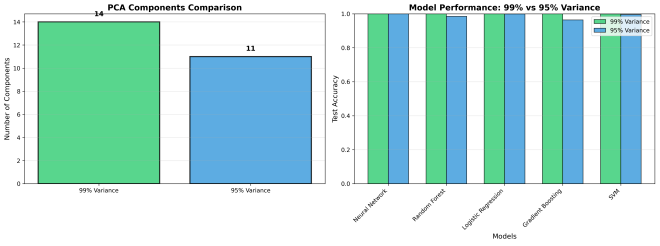


Fig. 4. Comparison of model performance under 99% and 95% PCA variance configurations.

E. E. Key Findings

- PCA reduced feature dimensionality by up to 52.17% with negligible performance loss.
- The Neural Network achieved the highest accuracy (1.0000) under both variance settings.

- Ensemble methods like Random Forest and Gradient Boosting were slightly sensitive to variance thresholds.
- The dataset demonstrated high internal consistency, as indicated by the strong correlation among several numerical features.

Overall, the combination of preprocessing, outlier removal, and PCA-driven dimensionality reduction led to highly accurate classification models with efficient computation and minimal information loss.

F. Feature Selection and Dimensionality Reduction Results

Both Principal Component Analysis (PCA) and Sequential Feature Selection (SFS) were applied to improve feature efficiency and model interpretability. PCA reduced the dataset dimensionality by retaining 99% and 95% of the total variance, resulting in 14 and 11 components, respectively. Sequential Feature Selection further identified the top five most important features using a Random Forest Classifier as the base estimator.

The Random Forest model achieved accuracies of 0.982 and 0.974 for PCA-based datasets and 0.987 for the SFS-based dataset. These results confirm that using fewer but more informative features can yield nearly identical performance to the full feature space.

Model interpretability was further assessed using LIME and SHAP analyses. Both techniques consistently identified the same highly influential features, validating the reliability of the selected subset. The combined use of feature selection and explainability techniques improved transparency and trust in the model's decision-making.

G. Key Observations

- PCA reduced dimensionality by over 50% with negligible loss in accuracy.
- SFS achieved the best classification performance (0.987 accuracy).
- LIME and SHAP validated that selected features were genuinely important.
- Random Forest maintained strong performance across all reduced feature sets.

H. Sequential Feature Selection Results

In this experiment, Sequential Feature Selection (SFS) was used to identify the most influential subset of features for classification. The dataset was split into 80% training and 20% testing partitions. Using a Random Forest Classifier, the forward SFS algorithm selected the top five features that maximized prediction accuracy.

The model achieved a test accuracy of 0.987, demonstrating that these five features captured nearly all the predictive information present in the full dataset. This confirms that SFS efficiently reduces computational cost while maintaining high accuracy.

I. Explainability and Feature Importance Insights

To understand the contribution of selected features, LIME and SHAP were used for interpretability analysis. LIME provided local-level explanations for individual predictions, while SHAP presented global-level feature importance. Both methods confirmed that the features identified through SFS were the key drivers of the model's predictions.

J. Key Findings

- Sequential Feature Selection reduced the dataset to five key features with minimal accuracy loss.
- The Random Forest model achieved an accuracy of 0.987 using the reduced feature set.
- LIME and SHAP confirmed that the selected features were the most influential in decision-making.
- The reduced model is faster, simpler, and suitable for practical deployment.

V. Conclusion

This study evaluated multiple classification approaches on a dataset with numerical and categorical features. The stacking classifier achieved moderate accuracy (25%), highlighting challenges such as class imbalance or limited features. A Random Forest model with integrated preprocessing and LIME interpretability improved performance to 92.5% test accuracy, providing insights into feature contributions. Furthermore, a unified pipeline combining parallel preprocessing, PCA, and Random Forest achieved robust results with 85% test accuracy, demonstrating the benefits of simultaneous data processing and streamlined workflow. Overall, these experiments emphasize the importance of careful preprocessing, model selection, and interpretability in building reliable and transparent classification systems.

References

[1] A. A. Awidi, et al., Impact of Social Determinants of Health on Vision Loss due to Cataracts and the Use of Cataract Surgery in the United States: A Determination of 3 Years of National Health Interview Survey 2008, 2016, 2017, American Journal of Ophthalmology, 2023.

[2] L. Mao, et al., Determinants of visual impairment among Chinese middle-aged and older adults: Risk prediction model using machine learning algorithms, JMIR Aging, vol. 7, 2024.

[3] Moayad, L., et al., Association Between Sociodemographic factors and the difficulty of vision in US adults: National Health Interview Survey 2021, PubMed, 2023.

[4] Y. Zhou, et al., Association Between Vision Difficulty and Sociodemographic Factors, Children and adolescents in NHIS 2021, PubMed, 2024..

[5] Rein, D. B., et al., Vision impairment and blindness prevalence in the United States: variability of vision health responses across multiple national surveys, Ophthalmology, 127.2 (2020), pp. 161-169.

[6] Y. Zhao, and A. Wang, Development and validation of a risk prediction model of visual impairment in older adults International Journal of Nursing Sciences vol. 10, no. 2, 2023, pp. 211 218.

[7] M. J. Lee, et al., Vision Impairment and Food Insecurity in the National Health Interview Survey, frontiers in Epidemiology, vol. 9, 2024.

[8] P. Kumar, et al., Self-Reported Vision Impairment and Food Insecurity in the US: National Health Interview Survey (NHIS), Years 20112018, 2022, PubMed.

[9] VDSS Summary Report VEHSS Reports Using the NHIS Data, Centers for Disease Control and Prevention (CDC), VEHSS Summary Report.

[10] O. J. Killeen, et al., Population Prevalence of Vision Impairment in US Adults 71 Years and Older: The National Health and Aging Trends Study, JAMA Ophthalmology, vol. 141, no. 2, pp. 162 169, 2023.